Status	Finished
Started	Wednesday, 21 October 2026, 10:28 PM
Completed	Wednesday, 21 October 2026, 10:44 PM
Duration	16 mins 6 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array

b. Item is the last element in the array or item is not there at all

c. When the item is somewhere in the middle of the array ✓

d. When the item is not the array at all

Question **2**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted

b. When the list is small and unsorted ✓

c. When the list is small and sorted

d. When the list is large and sorted

Question **3**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining

b. Finding

c. Discovering

d. Searching ✓

Question **4**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search continues in the right sublist
c. The search ends successfully ✓
d. The list is sorted

Question **5**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search works on unsorted lists
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search can find multiple instances of the target element

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted
b. Number values should only be present

Number values should only be present

c. There should be direct access to the middle element in any sublist
d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search
b. Both (1) & (3)
c. Linear search ✓
d. Binary search

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O👎
b. O(n log n)
c. O(1)
d. O(log n) ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search
b. Hashing
c. Linear search ✓
d. Binary search

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
b. There must be a mechanism to access middle element directly
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. Must use a sorted array

Question **11**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list is sorted
b. When the search algorithm finishes
c. When the list contains duplicate elements
d. When the element is found in the list ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The list is sorted
d. The search continues from the beginning

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list contains duplicate elements
c. When the list is very small
d. When the list is unsorted

Question **14**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list ✓
b. Divide the list into two equal parts
c. Compare the target element with the first element in the list
d. Sort the list

Question **15**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(1) ✓
b. O👎
c. O(n log n)
d. O(log n)